

Activity 1: The Library System (String Functions)

Focus: UPPER(), LOWER(), LENGTH(), SUBSTRING() / LEFT()

Setup Code (Tell students to run this first):

```
SQL
CREATE DATABASE FunctionDB;
USE FunctionDB;

CREATE TABLE Books (
    BookID INT PRIMARY KEY,
    Title VARCHAR(100),
    Author VARCHAR(50)
);

INSERT INTO Books (BookID, Title, Author) VALUES
(1, 'Harry Potter', 'J.K. Rowling'),
(2, 'The Hobbit', 'J.R.R. Tolkien'),
(3, 'game of thrones', 'George R.R. Martin');
```

Questions for Students:

1. Write a query to display all Book Titles in **UPPERCASE**.
2. Write a query to display the Author's name and the **number of characters** (length) in their name.
3. We need to create a short code for books. Write a query to display the **first 3 letters** of the Book Title. (Hint: Use LEFT() or SUBSTRING())

Activity 2: Student Exam Results (Math Functions)

Focus: ROUND(), CEIL(), FLOOR()

Setup Code:

```
SQL
CREATE TABLE ExamResults (
    StudentID INT PRIMARY KEY,
    StudentName VARCHAR(50),
    Marks DECIMAL(5,2)
);

INSERT INTO ExamResults (StudentID, StudentName, Marks) VALUES
(101, 'Nimal', 85.75),
(102, 'Kamal', 92.40),
(103, 'Sunil', 74.99);
```

Questions for Students:

1. The school wants to round the marks to the nearest whole number. Write a query using the **ROUND()** function to display the StudentName and their rounded marks.
2. The science teacher wants to give extra marks by rounding UP everyone's marks to the next highest whole number. Write a query using the **CEIL()** function.
3. The strict math teacher wants to round DOWN everyone's marks to the lowest whole number. Write a query using the **FLOOR()** function.

Activity 3: Employee Leave Tracker (Date Functions)

Focus: YEAR(), MONTH(), DATEDIFF()

Setup Code:

```
SQL
CREATE TABLE EmployeeLeaves (
    LeaveID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    StartDate DATE,
    EndDate DATE
);

INSERT INTO EmployeeLeaves (LeaveID, EmpName, StartDate, EndDate) VALUES
(1, 'Amala', '2023-10-01', '2023-10-05'),
(2, 'Ruwan', '2023-12-15', '2023-12-20'),
(3, 'Kasun', '2024-01-10', '2024-01-12');
```

Questions for Students:

1. Write a query to display the EmpName and only the **Year** they started their leave.
2. Write a query to display the EmpName and only the **Month** their leave ended.
3. Calculate the total number of days each employee was on leave. Write a query using the **DATEDIFF()** function to find the difference between EndDate and StartDate.

Activity 4: Shop Sales Summary (Aggregate Functions)

Focus: SUM(), AVG(), MAX(), MIN(), COUNT()

Setup Code:

```
SQL
CREATE TABLE DailySales (
    SaleID INT PRIMARY KEY,
    ItemName VARCHAR(50),
    Amount DECIMAL(10,2)
);

INSERT INTO DailySales (SaleID, ItemName, Amount) VALUES
(1, 'Laptop', 150000.00),
(2, 'Mouse', 2500.00),
(3, 'Keyboard', 4500.00),
(4, 'Monitor', 35000.00),
(5, 'USB Drive', 1500.00);
```

Questions for Students:

1. The shop owner wants to know the **Total** sales amount for the day. Write a query using SUM().
2. Find the **Average** (AVG) price of the items sold today.
3. Write a single query to find both the **Highest** (MAX) sale amount and the **Lowest** (MIN) sale amount.
4. Count how many total items (transactions) were sold today using the COUNT() function.